Lab 1: Requirements Specification Table

Problem Statement #42: Internal Microservice Catalog & Health Portal | SE Lab (CSE 5A)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	High	Pass: Downtime recorded and alert dispatched on 3 consecutive failed health probes. Fail: Unreachable service marked healthy.	Core health monitoring functionality to ensure timely detection of service outages.	Baseline requirement from problem statement.
FR-002	Functional	The system shall allow a DevOps Engineer to register a new microservice by specifying its name, owner team, repo URL, API doc endpoint, and health check URL.	High	Pass: Service entry added to catalog, API endpoint validated, initial probe runs in < 5s. Fail: Invalid or unreachable URL registered without returning error.	Required for populating and keeping the enterprise service catalog up to date.	Input validation required on URLs.
FR-003	Functional	The system shall visualize microservice dependencies as an interactive dependency graph mapping upstream and downstream service connections.	High	Pass: Graph accurately renders all service nodes/edges and supports node filtering. Fail: Circular dependencies cause visual graph render crash.	Enables System Architects to analyze topology and assess failure blast radius.	Node graph should support pan/zoom.
FR-004	Functional	The system shall dispatch real-time downtime notifications to DevOps Engineers via email and Slack webhook when a service health status changes to 'Unhealthy'.	Medium	Pass: Alert delivered to Slack channel and email within 10s of detected downtime. Fail: Alert lost or delayed beyond 30s.	Reduces Mean Time to Detection (MTTD) for operational incidents.	Include rate-limiting to prevent spam.
FR-005	Functional	The system shall aggregate and display searchable OpenAPI/Swagger documentation for all registered microservices.	Medium	Pass: Developers can search by service name or tag and view rendered API endpoints. Fail: API docs fail to render or show stale schema.	Facilitates microservice discovery and API reuse across dev teams.	Support OpenAPI 3.0 specs.
NFR-001	Non-Func (Performance)	The service catalog dependency graph viewer must render interactive node architectures containing up to 200 services smoothly without UI degradation.	High	Pass: Benchmarks confirm latency < 2.0s and 60 FPS interactions with 200 nodes & 500 edges. Fail: Render time > 3.0s or frame drops below 30 FPS.	Ensures System Architects can navigate large service mesh topologies effectively.	Baseline requirement from problem statement.
NFR-002	Non-Func (Security)	The system shall enforce Role-Based Access Control (RBAC) via enterprise OAuth2/SSO and maintain 99.9% portal uptime during operational hours.	High	Pass: Unauthorized write attempts blocked with HTTP 403, portal uptime >= 99.9% over 30 days. Fail: Unauthorized catalog edits or uptime drops < 99.9%.	Protects service metadata integrity and guarantees high operational availability.	Integrate with enterprise SSO (Okta).